



PTT Global Chemical Public Company Limited

GC Management System and Process Safety

SG-(Q-MP)-014

Guidance for Hazard and Operability Studies (HAZOP)

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Edition records

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1	01/03/2025	Create New Document	Miss Chaowarat Bunditkamol
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Related Units

Unit Code	Unit Name
Q-MP-TS	Technical Safety Service

Related KPI

KPI Measure	Description / Calculation	Target (unit)

Related Law

Law Name

Related Documents

Document ID	Document Name
P-(Q-MP)-OEMS-005	Hazard and Operability Study (HAZOP)
P-(TP-PM)-OEMS-002	Management of Change (MoC) Procedure
SG-(Q-MP)-004	Guideline for Plant PSM Governance
SG-(Q-MP-PS)-001	Process Safety Management Leading Indicators

External Reference Documents

Document Name
Table A6.2-2 PSI readiness checklist (Final R1)
Table A6.2-3 PID readiness checklist (Final R1)

Internal Use Only

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1. Purpose/Objective

HAZOP is a systematic methodology to identify hazards and operability problems which is the one of techniques for **Process Hazards Analysis (PHA)** as mention in the **Process Safety Management (PSM)** regulation of **Industrial Estate Authority of Thailand (IEAT)** and is generally accepted as the preferred methodology in the chemical and petroleum industries. The concept involves investigating how the plant might deviate from the design intent.

The objectives of HAZOP study are to:

- Identify hazards and operability problems;
- Assess the extent of the hazard or operability problems;
- Identify the safeguard for each identified hazardous event;
- Propose recommendations to prevent or mitigate the consequence; and
- Serve as input to risk management activities

HAZOP methodology limitation

- HAZOP is not suitable for examining the high-level issues such as the details of layout or the quantification of risks that would be performed with more detailed assessments such as a **Quantitative Risk Assessment (QRA)**, a detailed facility siting study etc. Separate analyses are required to address the above issues properly.
- HAZOP is not the means for defining engineering and procedural solutions for sources of hazards.
- HAZOP relies critically on expertise and knowledge of the team members involved.
- HAZOP quality is affected by the accuracy of **Process Safety Information (PSI)**. Inaccurate **PSI** such as P&IDs, information on process equipment, materials of construction, chemical information, etc. will result in additional time required and could lead the HAZOP team to flawed conclusions

The aim of this guideline is to ensure that the HAZOP study in PTT Global Chemical Public Company Limited (hereinafter referred to as “**GC**”) is conducted by standardize

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2. Scope

- The HAZOP methodology and requirement in this guideline shall be applied when the HAZOP technique is selected to perform the process hazard analysis.
- If *Term of reference (TOR)* is required, the requirement and HAZOP Methodology as provided in this guideline shall be applied to prepare the HAZOP TOR and conduct the HAZOP study.
- If the HAZOP is performed under scope of the Hazard and Operability Studies (HAZOP) P-(Q-MP)-001-OEMS-005 (Called *HAZOP Procedure*), the detailed narrative workflow as provide in *HAZOP Procedure* shall be followed. The HAZOP methodology in this guideline shall be applied during perform HAZOP study.
- If legislative or other governmental requirements conflict with the methodology in this guideline, governmental requirements shall prevail and be complied, and If local codes or practices conflict with this guideline, the more stringent requirement shall be adopted

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3. Roles and Responsibility

3.1 HAZOP Guideline owner

HAZOP guideline owner is Technical Safety Services (Q-MP-TS) Division Manager which has the responsibility to

- Approve/Review the HAZOP study guideline when the document is revised and oversee effective of **HAZOP**.

3.2 HAZOP Guideline custodian

Technical Safety Services (Q-MP-TS) engineer is the HAZOP guideline custodian who has responsible for

- Issue the HAZOP study guideline;
- Review and identify the opportunities for improvement of HAZOP methodology;
- Certify and record the internal and external HAZOP Leader;
- Ensure that this guideline is implemented effectively and communicated to all relevant users.

3.3 HAZOP Requester

HAZOP requester is the person that requests to perform the HAZOP study which may be:

- Project Director/Manager in case the HAZOP is performed under the scope of project;
- **MoC Champion** in case the HAZOP is performed under the scope of **MoC**. For this case **MoC champion** may take the role for both of HAZOP requester and HAZOP coordinator; or
- Plant Division Manager in case the HAZOP is performed for any issue (e.g. safety concern of operation, etc..) of the operating facility which is not under the scope of **MoC** or Project.

The HAZOP requester shall have responsible tasks including, but not limited to;

- Consider the necessities of HAZOP study;
- Ensure that **HAZOP TOR** or HAZOP study session is conducted as the requirements of this guideline;
- Provide the budget and resource for HAZOP study including identifying the representative who is the HAZOP coordinator;

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- Approve the HAZOP study report including HAZOP Recommendation;
- Ensure the changing of Recommendation and Action are reviewed by relevant parties;
and
- Ensure that the recommendations are completely closed and implemented in timely manner.

3.4 HAZOP Coordinator

HAZOP coordinator is the person who response to coordinate the HAZOP study which may be:

- Project Director/Manager or Project engineer or representative person who is assigned by Project Director/Manager in case of the HAZOP is conducted by project;
- **MoC Champion** in case the HAZOP is performed under the scope of **MoC**. For this case, **MoC champion** may take the role both of HAZOP coordinator and HAZOP requestor;
- Plant Division Manager or representative person in case the HAZOP is performed for any issue (e.g. safety concern of operation, etc.) of the operating facility which not under the scope of **MoC** or Project

HAZOP Coordinator shall have responsible tasks follow **HAZOP procedure**.

- Define the scope and objectives of the study.
- Planning HAZOP schedule and Coordinate the **HAZOP study**.
- Appoint a HAZOP Leader, HAZOP team members, and HAZOP scribe.
- Ensure that HAZOP Leader has the qualification as requirement.
- Prepare **Process Safety Information (PSI)** for HAZOP team and ensure that necessary information is distributed to HAZOP team members (at least 3 days prior to HAZOP Workshop start is preferred).
- **Ensure that **PSI** is available and accurate for HAZOP study.**
- **Complete the PSI readiness checklist (Including P&ID readiness checklist) and send evaluation results to HAZOP Leader before conducting study.**
- Arrange a HAZOP study.
- Ensure that all HAZOP recommendation items from HAZOP study are tracked and implemented properly.
- Raise the **HAZOP action change** and get agreement from HAZOP team when **HAZOP action change** is required.

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- Ensure that the most updated **HAZOP study** report is registered (For HAZOP under scope of **MoC**, HAZOP study report shall register in the **e-PHA** system.)
- Monitor the HAZOP schedule and update to ensure that HAZOP workshop is completed as planned.

3.5 HAZOP Leader

HAZOP Leader is **GC** staff or qualified Contractor who has experience and is qualified as the requirement of this procedure.

HAZOP Leader shall have responsible tasks follow **HAZOP procedure**.

- Review the PSI readiness checklist.
- Advising project/site leadership of the need to delay/postpone the study until issues affecting the integrity of the HAZOP can be resolved.
- Gather information for HAZOP node preparation for the **HAZOP study**.
- Review the design to determine sufficient technical information and required skills of study team members.
- Verify adequacy of team member to ensure effectiveness of the study meeting with multi-disciplinary team and using “brainstorms” approach.
- Facilitate the meeting/workshop and ensure that all issues are covered in adequate depth.
- Being alert to time pressures and ensure that it does not compromise the quality, thoroughness, or integrity of the review.
- Review and approve **HAZOP study** report.

3.6 HAZOP Scribe

HAZOP Scribe shall be the person who has the strong knowledge about HAZOP study. Each HAZOP study should have a dedicated HAZOP scribe which is not involved in brainstorming to increase the effectiveness of recording.

HAZOP scribe shall have responsible tasks follow **HAZOP procedure**:

- Record identified hazards and problem areas and capable of structuring recommendations/ actions in a clear and understandable way.
- Complete the HAZOP report and send the final draft to HAZOP Leader for approval.

3.7 Responsible person

The Responsible person shall be the person who is assigned by the HAZOP team.

Responsible person shall have responsible tasks follow **HAZOP procedure**.

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- Implement or/and follow up the action item and close action item as agreed by the HAZOP team. (For HAZOP under scopr of **MoC**, action item shall be closed and recorded in the **e-PHA** system.)

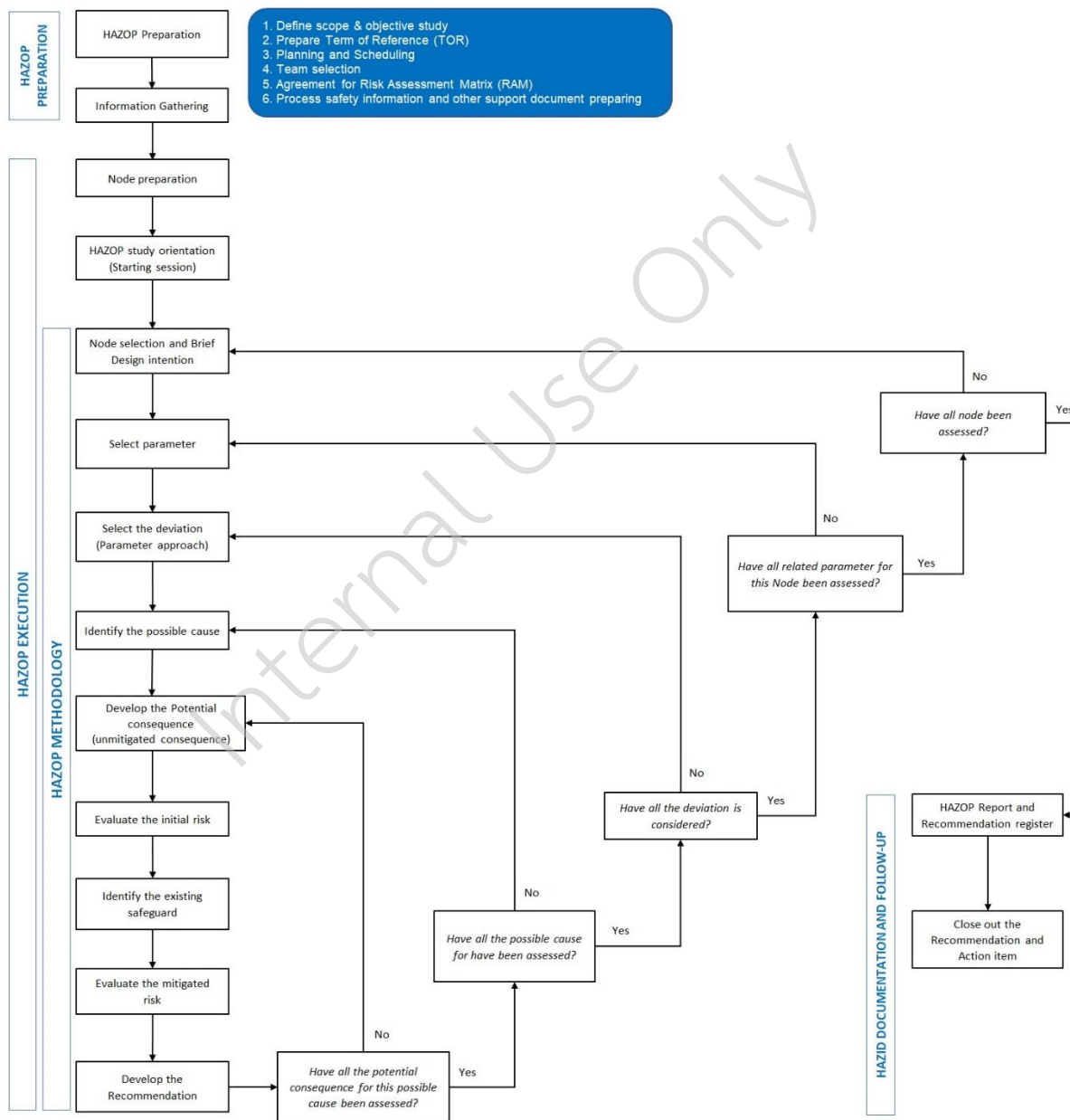
3.8 Action item approver

Action item approver shall be the person who is assigned by HAZOP team. (If there is no specific person as team consensus, HAZOP requester or the person who is assign by HAZOP requester shall be the action approver)

Action items approver shall have responsible tasks follow **HAZOP procedure**.

- Action item approver is the person who has been assigned to review, approve or reject proposed closed-out action item. If there is no specific person as team consensus, Project manager (For project does not cover by **MoC**) or **MoC champion** (For project under **MoC**) shall be the action approver.

4. Workflow



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5. Detailed Narrative of Workflow

5.1 HAZOP application in facility life cycle

Project phase – Select phase.

- During **SELECT** phase, the Authorized Person in consultation with the Sponsor (e.g. Project manager or Project representative who is assigned by project manager) or **MoC champion** shall determine if a coarse HAZOP is relevant and appropriate to the project.
- If the HAZOP is required the HAZOP study should be performed for all alternatives before the next phase, in order to be the support information for technology, process or system selection and serve as the information for inherently safer design prior to **Front End Engineering Design (FEED)** in **DEFINE AND DEVELOP** phase

Project phase - Define and Develop phase.

- HAZOP in **DEFINE AND DEVELOP** phase shall be conducted with complete documentation more than the previous phase as describe below.
 - a) The design intent, mode of operation, operating condition, process step and safeguarding instrumentation shall be available for team understanding.
 - b) P&ID and PFD shall be approved for HAZOP. The version shall be at least client comment version and include the comment from review session.
 - c) P&ID shall show all equipment, piping class and set pressure. In case there is package unit(s) in scope of review, P&ID of the package unit(s) shall be available to provide enough information.

Note: HAZOP study shall not be the session for P&ID review. Therefore, session for P&ID review shall be arranged prior to the HAZOP study



- d) In case of project interfaces with existing facility, the “as built” P&IDs for the existing installation shall be used. P&ID shall show the tie-in interface between new facility/modification and existing facility.
 - e) The equipment and instrument list shall be available to show information that may not be provided in P&ID (e.g., sizes, design temperatures, pressures and materials).
 - f) The document required for HAZOP in this phase are referred Appendix 6.2
- HAZOP team shall review the HAZOP Recommendations from **SELECT** phase including the other safety studies such as the **Hazard Identification (HAZID)** study from the previous project phase.
 - Project manager or project representative who is assigned by project manager or **MoC champion** shall determine the necessary to revisit, redo, or additional HAZOP study when there is/are change after the HAZOP for **FEED** is complete.

Project phase - Execute phase.

- HAZOP in EXECUTE phase shall be completed before construction activities. This HAZOP shall be performed based on the complete detail of engineering documents which were stamped as “Approved for Construction”.
- All HAZOP recommendations from the **SELECT** and **DEFINE & DEVELOP** phases shall be reviewed to ensure all recommendations are closed out and used as information.
- HAZOP in **EXECUTE** phase shall be conducted by following the full HAZOP methodology as described in this guideline, except the study is performed to focus on changes since previous HAZOP report in **DEFINE & DEVELOP** phases.
- The required documents for HAZOP during this phase are referred Appendix 6.2

- Project manager or project representative who is assigned by project manager, or **MoC champion** shall determine the necessary to revisit, redo, or additional HAZOP study when there is/are change after the HAZOP for construction is complete (e.g., change in process condition, change the equipment design / type, change that effect to operation step, Change process technology and etc.)

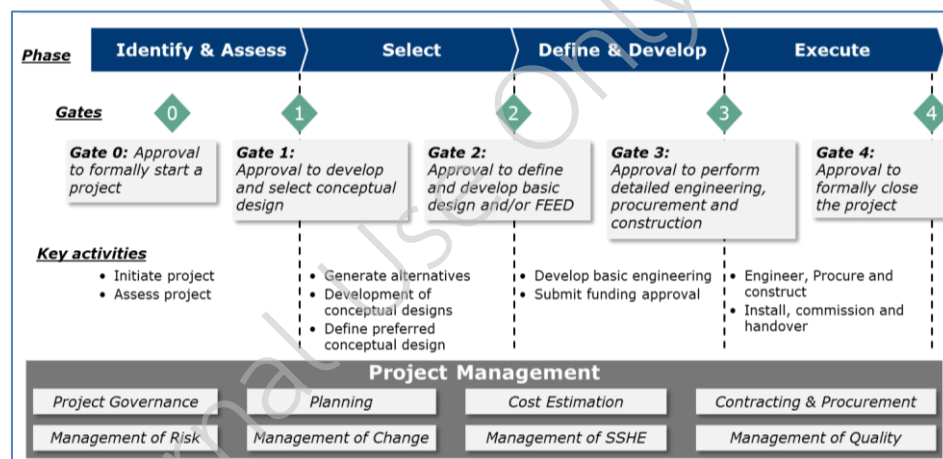


Figure 5.1 Gates of capital project management process

Operation phase

- Operating facilities should establish a schedule for completing or revalidating their HAZOP based on the requirements of Process Safety Management regulation and P-(Q-MP)-049 Process Safety Management. If the **HAZOP** is used to conduct the hazard evaluation, the HAZOP methodology as described in this guideline shall be used as the good practice to conduct the study.
- As changes take place, an MOC process as described in P-(TP-PM)-OEMS-002 Management of Change (Called **MoC procedure**) shall be followed. **HAZOP** can be used for hazard evaluation under the process of **MoC**, so that the hazards associated with those changes can be understood, documented, and mitigated.

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- The document required for **HAZOP** in operation phase are refer Appendix 6.2 (for facility not undergoing the modification. For the facility undergoing the modification the completeness of information may depend on the project phase or MoC process)

Non-Operation phase (Persevered and Abandoned or Decommissioning)

- As changes take place in the case of non-operation, an MOC process as described in **MoC procedure** shall be followed. **HAZOP** can be use ford hazard evaluation under the process of **MoC**, so that the hazards associated with those changes can be understood, documented, and mitigated such as preservation and demolishing especially the change to non-operation for some of equipment or system which is the one part of the main facility.
- The document required for **HAZOP** in non-operation phase are refer Appendix 6.2 (for facility not undergoing the modification. For the facility undergoing the modification the completeness of information may depend on the project phase or MoC process)

5.2 Planning and Preparation for HAZOP study

5.2.1 Term of reference (TOR)

- In case of HAZOP study is provided by the vendor, which requires **TOR**, the content shall be prepared based on this guideline and shall be reviewed/agreed between HAZOP requestor, HAZOP provider (vendor) and HAZOP guideline custodian. Generally, the content should contain the list below, but not limited to.
 - a) Project background
 - b) Objective of study
 - c) Methodology
 - d) Scope of work
 - e) Document for reviews
 - f) Activities and schedule
 - g) Venue

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- h) Deliverables
- i) Resources
- j) Other reference document

5.2.2 Planning and Scheduling

- HAZOP coordinator shall define the objective and scope of study.
- Schedule shall show the times (date and time) and duration of study sessions including dates on which draft and final reports are to be submitted to the various recipients.
- Schedule should consider the time required for a comprehensive review, the needs of the HAZOP team and availability of information and key team member required

5.2.3 Study room and location

- Study location should be selected based on location of design information, team members, external member or the facility to be reviewed
- The study room for HAZOP study should be selected by considering the sufficient size to comfortably accommodate the study team and any specialist advisors with enough working table space for placing copies of documents (e.g., PFD, Plot plan, P&IDs, etc.) for each team member, ventilation with adequate lighting, available of projector equipment or other support equipment

5.2.4 Agreement for Risk Assessment Matrix (RAM)

- **Risk Assessment Matrix (RAM)** for HAZOP study shall be agreed in prior perform study. **Process Hazard Analysis RAM (GC PHA RAM)** as provided in the Operational Risk Assessment Matrix and Its Applications W-(Q-MP)-002 shall be used for HAZOP study as the 1st priority. Using of other **RAM**, shall be agreed with owner of Operational Risk Assessment Matrix and Its Applications W-(Q-MP)-002 (If **TOR** is performed, agreed **RAM** shall be include in **TOR**)

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5.2.5 Team selection

HAZOP Team comprises of assigned representatives from concerned parties. The team should be as small as possible for each review session, preferably, no larger than 10 people and core team should not be changed during the HAZOP study and frequent change of HAZOP team during the study is not acceptable.

- Core team shall be multidiscipline and shall include full-time participation of following person

HAZOP Leader

HAZOP Leader is the key person to the success of the study. HAZOP Leader shall have the qualification as described below.

Internal HAZOP Leader shall have the qualification as described in **HAZOP Procedure**

External HAZOP Leader shall be the person who is qualified by HAZOP guideline custodian and HAZOP requestor. HAZOP coordinator / HAZOP requestor shall submit the F-(Q-MP-TS)-002 External HAZOP Leader Evaluation Form with curriculum vitae (CV) of external HAZOP leader HAZOP guideline custodian for approval and register as the qualified HAZOP leader is **e-PHA** system. To qualify the External HAZOP Leader, the following requirement shall be considered;

- Shall has the In-depth knowledge and experience in the HAZOP methodology and the concept of process hazard analysis (e.g., PHA training and PHA leader training);
- Shall has the experience for HAZOP study as the role of member or scribe.
- Shall has experience in Oil & Gas or Petrochemical industry more than 8 years.

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- d) Shall be the independent person from the design development, engineering contractor, and the facility being reviewed.

HAZOP scribe

HAZOP scribe has no other role, as recording is a full-time task separately with person involved in the brainstorming. To increase the effective of recording and brainstorming, the member of HAZOP team or HAZOP Leader shall not take the role HAZOP scribe. A competent HAZOP scribe should have the following.

- a) At least 2 years experience in Oil & Gas Industry or Petrochemical or Technical Safety Section.
- b) Must be familiar with the HAZOP study method and terminology
- c) Know how to use the software or HAZOP template used to record the study, have good typing and summarization skills

HAZOP member

HAZOP members shall consist of the multi-disciplinary team from relevant Departments. At the minimum, typical member for HAZOP team shall have representatives of Process engineer, Operation supervisor (at least), Maintenance engineer, Safety engineer and may include with the Project engineer in case of the HAZOP is conducted in scope of Project or MoC. HAZOP coordinator shall ensure that team consists of the following competent.

- a) Good knowledge of the design basis, design intention, operating condition and technology involved in the operation of the process as well as the hazards of the process.
- b) Good knowledge of the design basis, principle & working and limitation of equipment or system involved in the operation of the process as well as current hands-on maintenance experience in the process or system. This experience

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involves knowing how the facilities are maintained as opposed to how they are intended to be maintained

- c) Good experience with day-to-day operations and work practice. This experience involves knowing how the facilities actually operated as opposed to how they are intended to operate and familiar with the design intent and operating, start-up, shut-down, emergency and commissioning procedures.
- d) Good knowledge of occupational safety health and environment
- Support team member shall be the team member that supports the study as required and may not full-time participation such as technical expertise or specialist (e.g. Rotating equipment engineer, Civil/ Structural/ Piping engineer, etc.)

5.2.6 Process Safety Information (PSI) and support document preparation

- **PSI** and the other support information shall be listed and prepared to distribute to HAZOP Leader and team before conducting the study for reviewing prior HAZOP study.
- The reviewing of **PSI** and other support information shall not be conducted during the HAZOP session. Typical list of documents is provided in Appendix 6.2.
 - a) The operating facility not undergoing modification, **PSI** shall be signed off to “As-built” revision.
 - b) The operating facility undergoing the modification or the new facility which is confirmed to include the scope of study, HAZOP Leader shall evaluate the adequacy and accuracy of the information to ensure that information have been sufficiently developed appropriately for an effective review
 - c) HAZOP Coordinator (or Project team) should conduct PSI readiness checklist to ensure that the accuracy and adequacy of PSI is sufficient for HAZOP study. Refer Appendix 6.2 (Table A6.2-2 and Table A6.2-3) in SG-(Q-MP)-014 Guidance for



Hazard and Operability Studies (HAZOP). Evaluation result shall be informed to HAZOP Leader before HAZOP study session starts (At least 3 days before HAZOP study session is recommended).

5.3 HAZOP Execution

5.3.1 Node Preparation

Node is the breaking of the overall complex process design into a number of simpler sections which are then individually reviewed.

HAZOP Leader shall define the node of study for the effective brainstorming and manageable quantities of information to be handled. These nodes shall be discussed before starting the study to agree with the HAZOP team.

Guidance practice

- Node boundary should be selected by the functionality/intent to ensure that the design intentions of each node can be easily and clearly understood which should be encountered of a major piece of equipment (e.g. each node may only have one major piece of equipment such as a vessel or a heat exchanger) or contain with the multi-equipment with same design intention.
- In some cases, although the group of equipment has the same function or design intent, however if the equipment has more complexity or potential confuse over which piece of process equipment is being discussed, split into more than one node should be considered
- Global nodes can be applied to represent the whole process or certain aspects of it. They serve several purposes.
 - a) Global nodes allow initiating events that affect more than one node, entire process or parts of it. For example, some external events such as flooding, or the global loss of utilities such as electric power.

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b) Specific issues that arise in more than one node / system such as facility siting, human factors, and common piping issues etc.

- Size of node should appropriate with the complexity and the hazard of process which not too small and too large. (The first node of study may be small node; it eases people understand the overall process of the HAZOP study)

Too small node: Although small node as line by line (e.g. line by line approach) may easy understand but too small node may extremely time consuming and resulted in extensive repetition of recorded data and may lead to extreme fatigue and loss of interest by HAZOP teams, resulting in low-efficiency HAZOPs

Too large node: Although large may save the time of study but large node with many different equipment function or design intention may hard to handle, lead to team confusing and potential miss identification for some of hazard

- For the system that has more than one operation mode (e.g., Regeneration, De-coke, Significant change in production type or spec), HAZOP should be studied for each operation mode.

Another operation mode may be re-studied by repeating the same node boundary or identifying as the different node separately (e.g., Process that use the different process piping route in each operating mode).

The following issue is the guidance to separate node study when the systems have more than one operation mode.

- a) Different service fluid in system which cause of chemical hazard is changed for each operation mode
- b) Different process conditions which cause the different process hazard. For some of system, the service fluid is not change but the process condition is changed causing the changing of hazard (e.g. operation above and below autoignition,

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operate above-below decomposition temperature, change in level of exothermic reaction when the temperature increase, etc.)

- c) Different flow direction / different line up / or equipment path for each operation mode. For this case, node should be defined for each operation mode

5.3.2 Starting session

Before start HAZOP study, the HAZOP study orientation shall be provided to HAZOP team to ensure that team members are at the same point of knowledge in respect to the study

Guidance practice

- HAZOP Leader shall lead the orientation with the following content but not limited to;
 - a) Introduction of HAZOP team members
 - b) Scope of study and expectations.
 - c) Schedule and timing
 - d) Ground rules for the study and expectations of team members
 - e) Short brief of HAZOP methodology base on methodology in this guideline
 - f) Overall Process description (with the assistance of the project, process engineer who has the well knowledge for the process design being reviewed) and facility lay-out (e.g., other process, plant, community, publish river, sea, etc.)
 - g) Node conceptual. (Typically, HAZOP Leader shall prepare node of study. However, before start the session node shall be reviewed and agreed with HAZOP team)
 - h) Assumption of review (If applicable)

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5.3.3 HAZOP Study Methodology

During a brainstorming exercise of the HAZOP study, the following steps below shall be followed.

Step 1 - Node selection and Design intention

HAZOP study shall start with node selection. HAZOP Leader and HAZOP Team shall discuss to agree the node boundary (which is mark-up node in P&ID), then briefing and record of the Design intention including operating condition and design condition.

Guidance practice

- Design intention should indicate the following detail.
 - a) Operational intent.
 - b) Direction of stream to understand where stream come from and go to. The type of equipment or equipment tag/name should be refer to identify the equipment
 - c) Normal operating condition of Temperature, Pressure (and Flow, Quantities, pH, etc. if important to the node). In case node have the multiple equipment or difficult to describe the operating condition in the same paragraph of design intention, the operating condition may describe separately.
 - d) Specific information related to operational intent such as Exothermic or Endothermic reaction etc.
 - e) Clear scope which is not included in the node such as vendor packages, etc.

Step 2 - Select the deviation.

Deviation is the combination of process parameter and guideword which indicate departure from the design intent. HAZOP leader should think about the characteristics of process during selecting the deviation, which may require application of a comprehensive and standardized

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list of guide words for various types process. The typical of deviation are provided in Appendix 6.3

Guidance practice

- Start with the process parameter and combine with the related guideword to select the deviation (the parameter-first approach). It's easier to understand than starting with guidewords.
- Repeat parameter until all related guidewords of selected parameter are reviewed. Move to the next parameter and back again to continuous to the remaining guideword should be avoided.
- The list of all deviation shall be applied to each node (Full record approach). If there is no cause for that deviation , it should be documented to indicate that the deviation was considered, but there were no issues of concern (See below section “step 3 - Identify the possible cause”) or record “Not applicable”, “N/A” for those deviations.

Step 3 - Identify the possible cause.

Possible cause is the reasons why deviations might occur. HAZOP team shall identify the possible cause related with the deviation being studied. HAZOP Leader shall manage the brainstorming to ensure that team not try to find the root cause or any solution in HAZOP study.

Guidance practice

- Possible cause shall be identified base on the node being studied, the possible cause from the outside node should not be considered except first and last node of study but the potential consequence from the possible cause may occur throughout the process outside node
- More than one possible cause may be identified in the one deviation. Therefore, these possible causes shall be recorded separately for the effectiveness of assessment

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- The causes of any previous incident's unit or the similar process or technology should be discussed in study. An external incident may be considered, if applicable.
- There are opportunities to identify the same cause under multiple deviations. As long as the consequences and safeguards are fully defined and documented, there is no requirement to document the details for the same cause in each of the deviations. (An example would be a valve closing could result in both no flow or low pressure or low level).
- In case HAZOP team evaluate that no credible cause can occur for the deviation being reviewed, HAZOP team should record the result of review such as "No credible cause was/were identified", etc.
- Practice for recording the possible cause.
 - a) All causes (including failure mode) should be identified and clearly stated and provide sufficient information to describe the ways in which each hazardous event can occur.
 - b) The cause should not be a restatement of deviations.
 - c) Possible cause should specifically define by using equipment tag. Other method may be used such as refer P&ID grid, refer prior equipment, refer destination equipment, if equipment is not available to referring

***Note** Therefore, too short or non-specific possible cause such as "Valve fail", "Filter Plug" or restatement of deviations such as High pressure is system shall be avoided to recording in worksheet. Better recording should be considered such as "Malfunction of 13TIC001 leading to fully closing of 13TV003" or "Filter S-1301 at upstream of E-1331 partially plug".*

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Step 4 - Develop the potential consequence.

Potential consequence is the hazardous outcome statement that may include describing of intermediate and final event (Including, physical injury or damage to the health of people or damage to property or the environment)

Guidance practice

- It is significant to develop the consequences without giving credit to the safeguards (assuming no safeguard) which is called “unmitigated consequence”. Safeguards are discussed and documented in the next step.
- When there are multiple consequences related to the cause, the team should discuss and record each of them separately for easily evaluating the severity and identifying safeguard(s) which are related.
- When the unmitigated consequence may occur throughout the process outside node being reviewed, HAZOP Leader must decide whether to include the consequences in the node or to refer the analysis to later or more suitable node. Whichever decision is made, it is important that the consequences outside the study node are fully covered.

Typically, thinking the consequence backward to the upstream may be more difficult than thinking forward to downstream or at the node being studied and may lead to frequently recording of repeat consequence & safeguard. Therefore, thinking the consequence forward downstream or at the node being studied is a preferable approach.

- In case HAZOP team evaluate that no potential consequence can occur, HAZOP team should record the result of review such as “No consequence was identified”, “No adverse consequence”, “No significant consequence”, etc.
- Practice for recording the potential consequence.
 - a) All potential consequences shall be recorded clearly stated and provide sufficient information to describe the hazardous outcome.



- b) Think through and document the “step by step” chain of consequence and using the linking word (Potential for this to cause that, and then this result in that, causing this or that, etc.) to describe the intermediate event link to final unmitigated consequence.

***Note** Therefore, too short or unclear describing of potential consequence such as “Reactor runaway”, “Loss of containment”, “Operating upset” etc. shall be avoided to recording in worksheet. Better recording should be considered such as “High temperature of feed leading to reaction runaway (Extreme exothermic reaction) causing high pressure & temperature inside reactor (R-110) resulting in reactor fail/rupture potential fire, explosion and multiple fatality.*

Step 5 - Evaluate the initial risk.

Initial risk is the risk without considering existing safeguards. HAZOP team shall evaluate the initial risk with the following step.

***Note** The criteria in **GC PHA RAM** shall be used to evaluate the risk level except that the HAZOP study is performed based on another **RAM** as agreed in **TOR**. In such cases, the criteria of **Likelihood** and **Severity**, Aspect of concern (e.g. people, environment, economic and social) or the criteria of acceptable risk level may deviate from statement in below.*

1. Determine the severity level of the unmitigated consequence on the people, environment, economic and social.
2. Determine the **Likelihood** of consequence without consideration of safeguard (unmitigated consequence frequency) based on the assumption that when the possible cause occurs, the unmitigated consequence will always present. Therefore, unmitigated consequence frequency can be determined by the frequency of possible cause. See further information in Appendix 6.4 and 6.4.2

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3. Evaluate the initial risk by multiply maximum severity level of unmitigated consequence with unmitigated consequence frequency, then record the risk level (e.g. Very low, Low, Medium, High or Extreme)

Step 6 - Identify the existing safeguard.

Safeguard is the engineered systems or administrative controls designed to prevent or mitigate the consequences. The **existing safeguard** is engineered systems or administrative controls which already exist in the current design or installations.

Guidance practice

- Safeguards are not restricted inside of the node.
- HAZOP leader should challenge the existence of existing safeguards especially the existing facility such as actual status in field, operation practice, consistent with the engineering document, etc.
- For existing installations, the existing safeguard shall be the safeguard that is present in the local and document e.g. (P&ID, other existing engineering document, current operating procedure or work instruction etc.)
- For new installations or facilities under modification, the existing safeguard shall be the safeguard that presents in the design basis/philosophy or the technology of licensor, PFD, P&ID or any engineering document depending on the project state.
- Safeguarding may be an active or passive safeguard. During study, the team should understand the characteristics of safeguards for effective safeguard identification. HAZOP Leader should lead team to discuss working of safeguard by consider **Detect, Decide, Act**.

For example, in case alarm is identified as the safeguard, HAZOP leader should challenge the team to understand What is detect parameter? What is setting value? How did the operator respond to this alarm? Has the document support the action? These are

examples of questions for brainstorming. **However, do not try to audit all evidence during HAZOP study.**

- Practice for recording the existing safeguard.
 - a) Equipment tag, Numbering or any specific code should be used to identify the safeguard.
 - b) Principal safeguards (engineering and administrative controls) should be recorded in the HAZOP worksheet referencing the appropriate equipment tags, provide clear information to understand What protect and How to protect and protective condition if applicable (e.g. set pressure of PSV, Alarm set point, etc.)
 - c) Records the prevention safeguard first and then, mitigation safeguard. HAZOP leader should lead team by applying protection layer's concept. The protection layers that prevent hazardous event or mitigate the consequences are described in Figure 5.2

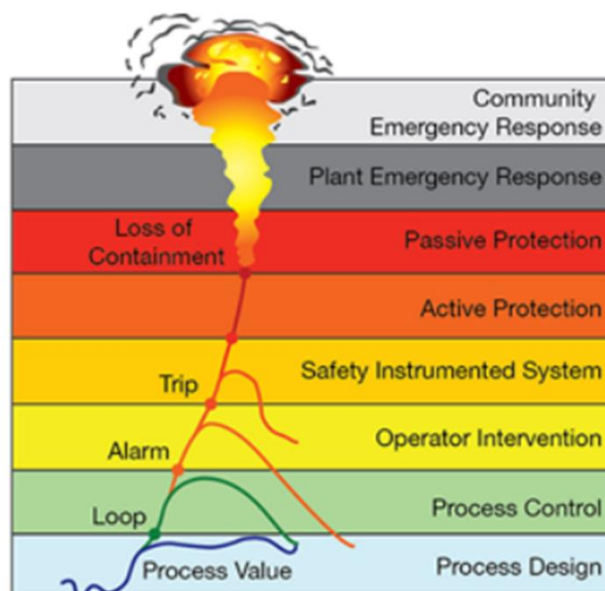


Figure 5.2 Examples of safeguards (Protection layers)

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***Note** General active fire protection and emergency response may be discussed in HAZOP study but are not listed as the safeguard (Not necessary to record in HAZOP), and not considered as the independent protection layers (IPL), as it is post event for most scenarios, and its availability and effectiveness may be affected by the several uncertain factor.*

Note Therefore, too short or unclear describing of existing safeguard such as “PSV”, “High level alarm”, “High pressure interlock” shall be avoided. Better recording should be considered such as “PSV-100 on HP separator V-1000 set pressure at 15 barg”, “Level high alarm LAH-100 on HP separator (V-1000) set at 65% with 10 min between LAH and full” or “Interlock Z-130 PAHH-001 (20 barg) to trip compressor (B-1201)”

Step 7 - Evaluate the mitigated risk.

Mitigated risk is the risk with considering of existing safeguard. The HAZOP team shall evaluate the mitigated risk by taking account credit of existing as following steps.

***Note** The criteria in **GC PHA RAM** shall be used to evaluate the risk except that the HAZOP study is performed based on other **RAM** as agreed in **TOR** such as joint venture project, the criteria of **Likelihood** and **Severity**, the aspect of concern (e.g. people, environment, economic and social) or the criteria of acceptable risk may deviate from statement in below.*

1. Determine the mitigated consequence frequency (**Likelihood** with safeguard) based on the identified existing safeguard.

Note The conceptual of Independent Protection Layers (IPLs) is recommended during determine the reduction magnitude order of **Likelihood**, if applicable. See further information in Appendix 6.4

2. Evaluate the initial risk (without safeguard) based on the result of multiplying between maximum severity level of unmitigated consequence and mitigated consequence frequency (**Likelihood** with safeguard). Record the risk level as the describe in **GC PHA RAM** or agreed **RAM** in **TOR**

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Step 8 - Develop the recommendation.

After mitigated risk is determined, HAZOP team may provide Recommendation, if the safeguards are considered insufficient which current mitigated risk is in unacceptable risk level (“Medium”, “High” or “Extreme” for **GC PHA RAM**).

Note Acceptable risk level for **GC** is “Low” or “Very Low”.

- When mitigated risk is in unacceptable risk level as agreed **RAM** of study (“Medium”, “High” or “Extreme” for **GC PHA RAM**), HAZOP team shall provide the Recommendation (one or more) to reduce the mitigated risk by proposing additional **IPLs**. Those Recommendation shall has the sum of IPLs credit (Risk reduction credit) sufficient to achieve acceptable risk (Acceptable risk level for **GC** is “Low” or “Very Low”) when those Recommendation are implemented.
- HAZOP team can provide the Recommendation also, although the mitigated risk already achieve the risk criteria as agreed **RAM** of study (“Low”, or “Very for **GC PHA RAM**”). These recommendations may be the additional **IPLs** to reduce the risk level or may be any improvement opportunities issue.
- Ensure that all recommendations have the responsible person for addressing and tracking progress of the corrective work until it is field verified that it was done including Action approver for approval of proposed action closed out.
- Recording of Recommendation
 - a) Recommendations should be thoroughly recorded so that a responsible person later can understand the recommendation, its intentions
 - b) Using the action word (Verb) such as Add, Change, Configure, Provide, Update, Prepare etc.
 - c) Explain the why the recommendation is needed, if possible

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Step 9 - HAZOP Worksheet Reference

Where causes, consequences or recommendations had been previously documented will be referred, HAZOP Leader and HAZOP scribe shall use a specific reference number.

- Do Not record by using “See above”, “See below”, “Same as above”, “Same as earlier node”
- Do not cascade the references.
- Do not refer to numbers in other nodes or other projects. It is better to copy contents from the other nodes to the node being studied.

5.4 HAZOP Documentation

5.4.1 HAZOP Worksheet

GC HAZOP worksheet as provided in *e-PHA* system shall be primary template to document the HAZOP study. If other template is used, the content shall be provided consistent structure to record the required content as describe in section 5.3.3 “HAZOP study Methodology”

5.4.2 HAZOP Report

Report shall be prepared and sent to review before submitting the complete revision. In case of the HAZOP report is perform by contactor for submission, the following are the general contents of HAZOP report but not limited to;

Section 1: Cover Sheet (form)

Section 2: Table of Contents

Section 3: Executive summary

Section 4: Abbreviations and Acronyms

Section 5: Reference

Section 6: Introduction

Section 7: Objective and Scope

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Section 8: HAZOP team

Section 9: HAZOP methodology and Assumptions

Section 10: HAZOP Basis & Study node list

Section 11: HAZOP summary of finding

Section 12: Appendices

Appendices section shall provide HAZOP worksheet and related document. The following are the content for appendices but not limited to;

- Team attendance for each session including the original sign in sheet.
- Marked drawings.
- TOR
- Risk Assessment Matrix
- HAZOP Worksheet
- Proposal memorandum for the HAZOP study or HAZOP Procedure including terms of reference, if applicable

5.5 HAZOP Recommendation Tracking and Follow – up.

5.5.1 HAZOP Report

- HAZOP Coordinator shall distribute / communicate the actions (action plan, action to be taken, completed date and responsible person) from **HAZOP study** to responsible person
- All recommendations shall be closed out by the responsible person and approved by Action item approver.

Note e-PHA shall be used for Tracking and Follow - up as the primary system.

However, where **e-PHA** is not available, a hard copy of recommendation sheet shall be generated for signing.

5.5.2 Modify Recommendation and Action item.

During action closing-out, Responsible person, HAZOP coordinator or any related person could possibly propose a resolution which may differ from the original recommendation, if

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the existing HAZOP recommendation is not possible / not practical to implement or the newly proposed action will provide safer solution. In such case, the **HAZOP action change** process as described below shall be followed.

- Proposed change shall be inform to HAZOP coordinator to evaluate the necessities of change
- The risk of hazardous events that relate to those recommendations shall be re-evaluated to ensure that the risk is at an acceptable level. Re-evaluation shall be performed by HAZOP team members (Original HAZOP team member should be involved).
- The agreed proposed change from re-evaluation shall be approved by the **HAZOP Requester** and shall have the record to indicate the approval.

Note If the recommendations are follow-up via **e-PHA** system, agreed proposed change and evident for approval shall be record in **e-PHA** system. In case the recommendation sheet is used, changing of action shall be recorded and sign in recommendation sheet.

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6. Appendix

6.1 Terms and Definitions

<i>Design intent</i>	Designer's desired, or specified range of behavior for parameters and characteristics
<i>Deviation</i>	The combination of parameter and guideword that help the HAZOP team investigate how the plant might deviate from the design intent.
<i>e-PHA</i>	internal GC electronic system to perform and record PHA report including follow up, tracking and monitoring of PHA recommendation
<i>Existing Safeguard</i>	Engineered systems or administrative controls which already exist in the current design or installations
<i>External HAZOP Leader</i>	External resource who is qualified to be HAZOP leader
<i>FEED</i>	Front End Engineering Design
<i>GC</i>	PTT Global Chemical Public Company Limited
<i>GC PHA RAM</i>	<i>GC</i> Risk Assessment Matrix for <i>PHA</i>
<i>Guideword</i>	Word or phrase which expresses and defines a specific type of deviation from a parameter's design intent.
<i>Harm</i>	Physical injury or damage to the health of people or damage to property or the environment
<i>HAZID</i>	One of Hazard Identification technique which is a qualitative technique for the early identification of potential hazards and threats effecting people, the environment, assets or reputation
<i>HAZOP</i>	Hazard and Operability Studies: A structured and systematic technique for examining a defined system, with the objective of identifying potential hazards and operability problems in the system.

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<i>HAZOP Action change</i>	The changing in detail of action or action cancellation which deviate from the original approved report
<i>IEAT</i>	Industrial Estate Authority of Thailand
<i>Initial risk</i>	The risk ranking that is determined based on the potential consequence and likelihood without consideration of the existing safeguard
<i>Internal HAZOP Leader</i>	GC staff who is qualified to be HAZOP leader
<i>IPL</i>	The protection layer from all of other protection layers and initiating event for given hazard which meet the criterial of independent protection layer
<i>Likelihood</i>	A measure of the expected probability or occurrence frequency of unmitigated consequence
<i>Mitigated risk</i>	The risk ranking that is determined based on the potential consequence and likelihood with consideration of the existing safeguard
<i>MoC</i>	Management of Plant Change (MoC) means management process for Changes which related to all Changes in Production Units.
<i>MoC Champion</i>	Anyone who is assigned to execute the Change. MOC Champion takes MOC through the entire process, ensuring that all appropriate reviews, procedures and documentation are completed during the various stages of the MOC. MOC Champion shall ensure that all the appropriate approvals are obtained for Implementation, Start-up and Close-out of the Change.
<i>Node</i>	A specific location in the process in which (the deviations of) the design/process intent are evaluated. It may be physical (e.g. hardware) or logical (e.g. step in an operational sequence).
<i>Parameter</i>	Variable that could cause deviations e.g. flow, pressure, temperature etc.
<i>PHA</i>	Process Hazards Analysis

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<i>Possible cause</i>	The reasons why deviations might occur
<i>Potential consequence (Unmitigated consequence)</i>	The hazardous event that might occur as the result of deviation without consideration of existing safeguard
<i>PSI</i>	Process Safety Information
<i>PSM</i>	Process Safety Management
<i>QRA</i>	Quantitative Risk Assessment
<i>RAM</i>	Risk Assessment Matrix: A matrix that is used during risk assessment to define the various levels of risk as the product of the harm severity categories and harm likelihood categories
<i>Risk</i>	Combination of the probability of occurrence of harm and the severity of that harm
<i>Severity</i>	A measure of the expected impact of hazardous event
<i>TOR</i>	Term of reference

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6.2 Readiness of Process Safety Information (PSI)

Table A6.2-1 Typical Process Safety Information (PSI)

Typical information required	Select	Define	Exceute	Operation	Non-operation
Operations and Maintenance philosophy	E	E	E	E	E
Process and Chemistry Descriptions	E	E	E	E	E
Process Control Narrative	D	E	E	E	E
Reactive Hazard (if applicable)	E	E	E	E	E
Material Safety Data Sheet	E	E	E	E	E
Plot Plan	E	E	E	E	E
General Equipment Arrangements & Elevations	-	E	E	E	E
Electrical/ Hazardous Area Classification Drawings	-	E	E	E	E
Heat and Mass balances	D	E	E	E	E
Process Flow Scheme (or PFD) and Utility Flow Scheme (UFS)	E	E	E	E	E
Process Engineering Flow Scheme and legend sheets (or P&IDs and UEFS)	D	D	E	E	E
Line lists for piping & process equipment	-	E	E	E	E
Equipment data sheet	-	D	E	E	E
Process Safeguarding Flow Scheme	D	E	E	E	E
Cause and Effect Diagrams	-	E	E	E	E
SIL Study report	-	-	E	E	E
Control valve turndown, RV, trip and alarm settings	-	E	E	E	E
Latest construction/ as-built drawings	-	-	E	E	E
Previous HAZID/HAZOP reports and Close Outs report	-	D	E	E	E
Details of change since last Process hazard analysis (PHA)	-	D	E	E	E
Other safety studies, such as Bow-tie analysis	-	-	D	E	E

Note E: Essential, D: Desirable

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Table A6.2-2 PSI Readiness checklist for HAZOP study

This template can download inform excel file. See attachment “External references” item

Table A6.2-2 Process Safety Information (PSI) readiness checklist for HAZOP Study						
	MoC / Project title :				Evaluation Date :	
Category	Example Critical Relate Document/Information	Review Issues (Accuracy / Completeness checks)	Review			Remark
			YES	NO	Not Related	
Chemical and Reaction hazard	GHS-compliant SDS <i>GHS: Globally Harmonized System of Classification and Labelling of Chemicals</i>	Safety Data Sheets (SDS) for all relate chemicals comply with GHS and are available for HAZOP when required checking the information. Focus section for HAZOP Section 2 Hazard identification Section 7 Handling and storage Section 9 Physical and chemical properties Section 10 Stability and reactivity				
Process Flow Information	- P&IDs, - PFDs with H&M balance	Latest revision of Related PFD, and/or P&IDs are available and approved for HAZOP.				
		PFD, and/or P&IDs of relate vendor package are available.				
		P&ID readiness checklist are available without deficiency or those deficiency can be solved such as correction with hand mark up? See Table A6.2-3	Refer result of Table A6.2-3 P&ID readiness checklist for HAZOP Study			
Process Description, Design Basis and Process chemistry	- Overall process description - Design intent - Operating envelope - Mode of Operation - Reaction information	Process description / Design intent for all mode of operation with clear understanding for operation scene.				
		Operation envelope (New/Updated/Current) for relate system and/or equipment are available. (Already check alignment with design basis documents)				
		Process chemistry information such as reaction information (may include side reaction) reaction rate, sensitivity, pressure and temperature profile etc are available				
Equipment information	- Equipment Data sheet	No differentiate from P&ID for existing facility				
		Relate vendor package data include equipment design (e.g., Design pressure and temperature) limit are available and must be aligned				
Safety Systems & Safeguards	- Cause and Effect diagram - SIS/SIF data/report	Information of new / existing relate alarm with updated set points are available				
		Cause and Effect information is available				
		Existing SIS/SIF information such as Design intent, SIL level (Target and Verification) are available.				
Relief & Flare Design	- Relief load calculation - Flare hydraulics - Dispersion study	Information for relief and vent system such as relief scenario and capacity is verified and available.				

Table A6.4 P&ID Readiness checklist for HAZOP study

This template can download inform excel file. See attachment “External references” item

Table A6.2-3 P&ID readiness checklist for HAZOP Study						
 MoC / Project title :		Evaluation Date :				
No.	Checklist Item	Purpose / What to Check	Review			Remark
			YES	NO	Not Related	
1	Latest revision issued & controlled Old revision used → missing design changes, leading to irrelevant or incorrect discussions.	Ensure HAZOP uses the most current approved revision.				
2	Scope completeness (systems, subsystems, boundaries) Parts of process omitted → hazards not identified.	Ensure that P&ID available for all relate scope of HAZOP. (PFD is not sufficient for HAZOP Study)				
3	Equipment tags complete & consistent Facilitator/team wastes time clarifying missing/duplicated equipment tags.	Ensures all equipment is identified - no missing of equipment - no incorrect equipment sequence - no duplicated tag / incorrect tag				
4	Line identification Inability to distinguish between critical and minor lines during hazard scenarios.	- All line complete with correct direction - Line identification complete (number, size, service, class etc.)				
5	Instrumentation (Indicators, transmitters, alarms) Risk of overlooking failure of unmarked instruments, incomplete/incorrect hazard identification, missing of barrier identification.	Ensure that all instrumentation present correctly - no missing instrument - correct instrumentation symbol - no missing/incorrect of alarm of alarm indication (e.g. L, LL, H, HH)				
6	Control loops & Interlocks Hazardous deviations missed leading to incomplete/incorrect hazard identification, missing of barrier identification and incorrect risk ranking.	Ensure that all control and interlock is present correctly. - no missing control / interlock loop - clear conception to the action element. Note Indication connection between sensor to final element may be not line-connection which depend on plant. - Fail position of valve				
7	Safety devices (PSVs, rupture discs, etc.) Relief scenarios cannot be validated; potential overpressure risk ignored.	Ensure that the relief device is present correctly. - no missing pressure relief device. - correct destination - set pressure is available correctly				
8	Tie-ins location Hazards at interface (wrong pressure/contamination) overlooked.	- Tie-ins location is identified - Isolation configuration follow reference specification				
9	Isolation devices shown (block valves, blinds) Potential misunderstand operation sence leading to improper/incorrect/incomplete develop consequence and Inability to evaluate isolation adequacy for maintenance/emergency	Block valves, Blinds, Indication for normal close and/or nomal open is present correctly and clearly understanding the operation sence.				
10	Consistency across multiple drawings Time wasted reconciling discrepancies between sheets.	Ensures that symbol for continuation drawing is present with correct detail.				
Conclusion by HAZOP Coordinator			YES	NO	Remark	
All item are available without deficiency or those deficiency can be solved such as correct/provide information with hand mark up?						

6.3 General HAZOP Deviation

The HAZOP team examines each *parameter* (and characteristic, where relevant) for *deviation* from the *design intent* which can lead to undesirable consequences. The identification of *deviations* from the *design intent* is achieved by a questioning process using predetermined “guide words”. The role of the guide word is to stimulate imaginative thinking, to focus the study and elicit ideas and discussion, thereby maximizing the chances of study completeness.

Basic guide words and *deviations* are shown below.

Table 6.1 List of typically Deviation

		Guideword					
		No/none	More/High	Less/low	Reverse	Other than	Change
Parameter	Flow	No Flow	More Flow	Less Flow	Reverse flow	Misdirected Flow	
	Pressure		High pressure	Low pressure			
	Temperature		High temperature	Low temperature			
	Level	No level	High level	Low level			
	Reaction	No reaction	Runaway	Low reaction	Reverse reaction	Side/ Wrong reaction	
	Mixing	No mixing	Excessive mixing	Low mixing			
	Phase						Phase change
	Viscosity		High viscosity	Low viscosity			
	Composition						Composition change



	Erosion/ Corrosion		High corrosion/ erosion				
	Service failures	Loss of service	Excess of service				
	Sequence					Wrong step	
	Incidents					Previous/ Related incident	
	Human factor					Human factor	

Table 6.2 Guidance for issue consideration in each deviation

Deviation	Consider	
No flow	- Block valve closed - Wrong line up - Slip blind installed - Equipment Failure (control valve, isolation valve, pump, vessel, instrumentation, etc.)	- Control valve fails closed - Incorrect pressure differential - Isolation in error - Power failure - Plugged line
More Flow	- Bypass valve open - Increased pumping capacity - Operation of pumps in parallel - Reduced delivery head. - Change in fluid density. - Exchanger tube leaks - Worn or deleted restriction orifice plates cross. - Connection of systems - Control faults	- Control valve trim changed. - Control valve fails to open. - Burst pipe. - Large leak - Wrong valve open - Wrong line up or misdirected flow - Slug flow - Water hammer - Increased flow from upstream process.
Less Flow	- Line restriction - Filter fouled - Defective pumps - Fouling of vessels - Lines - Valves or orifice plates	- Density or viscosity changes - Competing pump heads and flows - Inadvertently throttled valve - Incorrect valve sizing, Limit by existing pipe size - Wrong line up - Surging



Reverse Flow	- Malfunctioning - Omitted - Incorrect differential pressure - Two way flow - Emergency venting/ wrong line-up or misdirected flow	- In line spare equipment - Connections to utilities: water, N2, flush systems
Misdirected Flow	- Misdirected Flow due to Valve open in error or passing allowing material to be routed to an unintended location	
High Pressure	- Design pressures - Specification of pipes, vessels, fittings, and instruments - Pressure range for abnormal operations - Surge problems - Leakage from interconnected high-pressure system (HP to LP interface) - Gas breakthrough (inadequate venting) - Inadequate or defective isolation procedures for relief valves - Thermal over pressure - Exchanger tube leak.	- Positive displacement pumps - Control valves failed (closed or open) - Increased centrifugal pump suction pressure – start-up of spare pump. - Failure of ejector - Eductor system - More reaction - Plugged pressure tap. - Obstructed relief - Pressure testing - Excessive heating
Low Pressure	- Generation of low pressure or vacuum condition – cooling - Condensation - Gas dissolving in liquid - Restricted pump - Compressor suction line	- Undetected leakage - Vessel drainage procedure - Blockage of blanket gas - Failure of vacuum relief - Inadequate NPSH



High Temperature	- Fire situation - Ambient conditions - Fouled or failed exchanger tubes - Cooling water failure - Air cooler malfunction - Defective control - Heater control failure - Internal fires - Reaction control failures - Heating	- Medium leak into process - Heat tracing/ regeneration - Decoking - Heats of reaction - Mixing - Reactor hotspots - Decomposition or run-away reaction, Absorption, or solution - Burn protection. - Abnormal operations
Low Temperature	- Cold weather operations - Ambient conditions - Reducing pressure - Fouled or failed exchanger tubes	- Loss of heating - De-pressuring liquefied gas. - Endothermic reaction
No Level or; Low Level	- Inlet flow stops. - Leak - Drain valve left open. - Outflow greater than inflow - Control valve malfunction - Faulty level measurement - Incorrect calibration	- Two phase flow - Plugged instrument taps. - Inadequate - Residence time - Inadequate mixing, excessive heating - Gas in liquid lines.
High Level	- Outlet isolated or blocked. - Inflow greater than outflow - Control failure - Faulty level measurement - Incorrect calibration - Filling operations - Liquid in vapor lines	- Vessel overflow - Deactivated level - Inadequate time to respond. - Interface level control - Phase inversion - Slug flow - Condensation

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No Reaction Low Reaction Incomplete reaction	- Wrong reactant mix – low/high temperature - insufficient catalyst – channeling	- Not suitable proportion of initiator and inhibitor - Not suitable proportion chemical injection
Runaway Reaction	- Wrong reactant mix – low/high temperature - incompatible chemical - side reactions	- Not suitable proportion of initiator and inhibitor - Not suitable proportion chemical injection
Reverse Reaction	- Wrong reactant mix –low/high temperature - insufficient catalyst - channeling	
Side Reaction	- Wrong reactant mix –low/high temperature - insufficient catalyst - channeling	
Wrong Reaction	- Wrong reactant mix –low/high temperature - insufficient catalyst - channeling	
No Mixing Low (Poor) Mixing	- Agitator set at wrong speed - drive stops - agitator blade drops off - coupling failure - poor mixing.	
Excessive Mixing	- Agitator set at wrong speed	
Phase Change	- Pump cavitation - Liquid leak to compressor	
High Viscosity	- Incorrect material or composition - Wax or cooling of oils- loss of heating	- High reaction (e.g., Polymerization) - High solids concentration - Settling of slurries
Low Viscosity	- Incorrect material or composition - Incorrect temperature - Solvent flushing	-



Composition Change	- Leaking valves - Leaking exchanger tubes - Changes to feed stock - Inadequate quality control - Process control upset reaction intermediates. - By-products - Improper mixing - Phase inversion	- Ingress of air, water, or rust - Identify nitrogen interfaces to process. - Inadvertent mixing - Interconnected systems (especially services, blanket systems) - Wrong additives or catalysts - Catalyst poisons - Preparation for shutdown and start-up operations - Solvent flushing
High Corrosion / Erosion	- Materials specification - Fluid velocities - Sour service - Coating applications	-
Loss of service	- Instrument air - Stream - Nitrogen - Cooling water - Hydraulic power - Electric power	- Water supply - Telecommunications - PLCs/computers - HVAC - Fire protection (detection and suppression)
Excess of service	- Instrument air - Stream - Nitrogen - Cooling water	-
Wrong step	- Purging - Flushing - Start-up - Normal shutdown - Emergency shutdown	- Emergency operations - Inspection of operating machines - Pressure testing - Regeneration



Previous/Related incident	- Incidences at similar processes - Review of any previous incident with having potential for catastrophic consequences including near misses	- Undocumented incidents - were hazards addressed by the incident investigation? - Recommendations from the incident investigation resolved/implemented?
Human factor	- Operation too long / short - Operation mistimed. - Operation in wrong direction - Operation too little/too much - Operation too fast/too slow - Misalign - Right operation on wrong object - Wrong operation on right object - Operation omitted	- Operation incomplete - Operation too early/too late - Check omitted. - Check incomplete. - Right check on wrong object - Wrong check on right object - Check too early / late. - Difficult to access



6.4 Guidance to evaluate risk ranking [Refer to 5.3.3 step 5 and step 7]

As section 5.3.3 step 5 and step 7, two categories of risk ranking are required for HAZOP study. The first is initial risk (ranking without safeguard) and the second is mitigated risk (ranking with safeguard). This appendix is the guidance for HAZOP team to evaluate the risk ranking.

6.4.1 Conceptual

All initiating events (e.g., the failure of device, system, human error, etc.) have their own frequency of occurrence. When it occurs, it could lead to hazardous events and could escalate to unexpected consequences.

To avoid those events, the barrier will be provided to reduce the occurrence probability of unexpected consequences.

- Preventive barrier: This type of barrier is designed to prevent hazardous events when the initiating event occurs.
- Mitigative barrier: This type of barrier is designed to mitigate impact after hazardous events occur to avoid the escalation of events to unexpected consequence.

Note

Generally, mitigative barrier is not often recorded as the safeguard in HAZOP except for specific scenarios. For example, secondary containment (e.g., Dike/Bund) may be listed as the safeguard for chemical spill to public river scenario but could not reduce the likelihood fire or explosion scenario.

Based on this concept, presenting of protection barrier can reduce the frequency of the unexpected consequence due to when the initiating event occur, the barrier will prevent or mitigate the unexpected consequence. If those barriers do not fail, the unexpected consequence will not occur.

However, the barrier has its own probability of failure on demand. Therefore, it is not possible to eliminate the risk of occurrence of unexpected consequence as the result of occurrence of

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initiating event but presenting of sufficient protection barrier can reduce the occurrence frequency of unexpected consequence to acceptable frequency.

Compare with the HAZOP terminology, the initiating event, hazardous event/ unexpected consequence and protection barrier will be identified and recorded in HAZOP study worksheet in term of possible cause, potential consequence and safeguard respectively

In the HAZOP study the possible cause will be identified then follow by the related potential consequence. Next the initial risk without consideration of existing safeguard will be performed to understand the inherent risk. After that the safeguard will be identified and then evaluated for mitigated risk.

Initial risk will be evaluated from the severity and likelihood of potential consequence (*Unmitigated consequence which is the likelihood without safeguard*). The likelihood of potential consequence for initial risk will be determined based on the frequency of possible causes due to it assumption that if there are no safeguards, the potential consequence will always occur when the possible cause occurs. The safeguard will be identified in the next step, then followed by mitigated risk evaluation.

***Note** The guidance to estimate the level of likelihood for each possible are provided in section 6.4.2*

Mitigated risk will be evaluated from the severity of potential consequence and its likelihood which shall consider the presenting of existing safeguard. As the result of existing safeguard, the likelihood of potential consequence will be reduced while the severity is still not change.

Note** HAZOP team can apply the conceptual of independent protection layer (IPL) to estimate reduction magnitude level of likelihood. The guidance to evaluate the IPL and estimated reduction likelihood order are provided in section **Error! Reference source not found.

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6.4.2 Evaluation likelihood of event for initial risk ranking

The likelihood of event for initial risk will be determined based on the frequency of possible cause due to in this step, the existing safeguard is not considered. Therefore, it assumes that the potential consequence will always occur when the possible cause occurs

To determine the likelihood of possible causes, HAZOP team should think about the frequency of possible causes occurring in the past. This consideration may include the frequency of occurring of possible causes in the similar condition (e.g., similar process fluid, operating condition, design and type of equipment, type of facility) to avoid underestimation of likelihood. Typically, the possible cause often is the general equipment failure which has the typical generic failure rate. The information in Table 6.3 can be use as the guidance to estimate the likelihood level as the result of occurring of possible cause.

***Note** The information in Table 6.3 are just the guidance to estimate likelihood level base on the generic failure data which can applied for the common failure. However, some of possible causes are specific events which do not have statistical information. Therefore, site specific information should be used to evaluate the likelihood level by using the likelihood definition in GC PHA RAM. The result of evaluation is still relied on brainstorming and team consensus.*

***Warning** the information in Table 6.3 just the guidance for HAZOP team which is used for HAZOP study and not allow to use for SIL assessment and other quantitative or semi-quantitative assessment.*

***Warning** The guidance of likelihood level in Table 6.3 can applied only the HAZOP study that applied GC PHA RAM for risk evaluation.*

Table 6.3 Indicative guidance for likelihood estimation as a result of possible cause (initiating event)

No.	Initiating event (IE)	Generic Frequency Range	Guidance of likelihood Level (Qualitative)
1	Basic process control system (BPCS) instrument loop	1 to 1/10 year	Level 4
2	Human Error		
	The routine task that is performed once per week or more often (Task $\geq$ once per week)	1 year	Level 5
	The task is performed between once per year and once per week (Once per year $\leq$ Task $<$ once per week)	1 to 1/10 year	Level 4
	The task is performed less often than once a year. (Task $<$ Once per year)	1/10 to 1/100 year	Level 3
3	Pressure Regulator (Self-Regulator)	1/10 to 1/50 year	Level 4
4	Pump/ Fan/ Compressor Failure	1 to 1/10 year	Level 4
5	Loss of Supply from all causes e.g., Upstream supply pump failure, accidental block in from external source (off plot)	1 to 1/10 year	Level 4
6	Cooling Water Failure	No typical PFD	Refer to the cause of the cooling failure
7	Site-Wide Loss of Power (Refer to PHA RAM Likelihood definition)		
	If no history of Site-Wide Loss of Power in location.	No typical PFD	Level 3

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No.	Initiating event (IE)	Generic Frequency Range	Guidance of likelihood Level (Qualitative)
	If there is history of Site-Wide Loss of Power in location	No typical PFD	Level 4
	If there is history of Site-Wide Loss of Power in location more than once per year in location	No typical PFD	Level 5
8	Heat Exchanger Shell & Tube		
	Tube leaks	1/30 to 1/100 year	Level 3
	Shell leaks	1/3 to 1/10 year	Level 4
9	Mechanical Failures Metallic and Non-Metallic		
	No moving parts – No vibration (non-corrosive service)	1/50 to 1/100 year	Level 3
	Operate with vibration	1 to 1/50 year	Level 4
10	Partial Plug gage of Mechanical Equipment		
	Control valve, Dirty fluid	1/3 year	Level 4
	Filter/Stainer	1/21 year	Level 4 or 3 ^{Note 1,2}
	Pipeline Mixer	1/11 year	Level 4
	Reboiler Tubes	1/17 year	Level 4

Note 1 For clean service, or too large line size & mesh, likelihood level 3 may be considered to applied.

Note 2 If partially plug of filter cause the long-term effect consequence, likelihood level 3 may be considered to applied due to when the initiating event (Possible cause) occur (here is “filter partially plug”), the unmitigated consequence does not always or immediately occur.

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6.4.3 Evaluation likelihood of event for mitigated risk ranking

The likelihood of event for initial risk will be determined based on the frequency of possible cause due to in this step, the existing safeguard is not considered. Therefore, it assumes that the potential consequence will always occur when the possible cause occurs.

Consideration of Independent Protection Layer (IPLs)

IPLs are the independent protection layer from all of other protection layers and initiating event for given hazard. There should be no common cause failure or common mode failure that results in loss of protection afforded by IPL and any other protection layer.

All IPLs are safeguards, but not all safeguards are IPLs. The safeguard that considered as the IPLs shall satisfy the following criteria to be considered valid.

- a) **Effective:** The safeguard is described as “effective”, if it can prevent the unmitigated consequence when it functions as designed. To consider the effective, IPL shall be designed solely to prevent unmitigated consequence of one potentially hazardous event. The following is key thinking to consider the effective safeguard. “Big enough”, “Fast enough” and “Strong enough”

The preventive safeguard is considered as “effective” if it performs the intended function when the demanded and to the standard intent, and it capable on its own of preventing a deviation from developing into the hazardous event.

The mitigative safeguard is considered “effective” if it is capable of completely mitigating the consequence of the hazardous event. In other words, it is capable on its own of preventing an escalation of hazardous events to the unmitigated consequence.

- b) **Independent:** The safeguard is described as “independent”, if it is independent from initiating event and from other protection layers associated with identified potentially hazardous event. Independence requires that performance shall not be affected by failure of another protection layer or by conditions that caused another protection layer to fail. Protection layer shall also be independent of initiating cause.

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- c) **Auditable:** The safeguard is described as “Auditable”, if it be capable of the being audited to check that they work

***Note** However, HAZOP study is not audit, HAZOP Facilitator shall compromise and manage session to avoid document audit in the HAZOP session.*

To estimate the reducing of likelihood level as a result of safeguard that is considered as IPLs, the information in Table 6.4 and Table 6.5 can be used as the guidance for evaluation.

***Warning** The information in Table 6.4 and Table 6.5 just the guidance for HAZOP team which is used for HAZOP study and not allow to use for SIL assessment and other quantitative or semi-quantitative assessment.*

***Warning** The guidance for reduction of likelihood level in Table 6.4 and Table 6.5 can applied only the HAZOP study that applied GC PHA RAM for risk evaluation.*

***Warning** Do not use the reduction order of likelihood in Table 6.4 and Table 6.5 without consideration of IPLs due to without consideration of IPLs may lead to counting of reduction order of non-IPLs which cause risk underestimation.*

Additionally, some of safeguards may not considered as IPLs. List of those safeguard and the reason are provided in Table 6.6

Table 6.4 Index of Indicative guidance for reduction likelihood level of Active – IPLs

No.	Independent Protection Layer (ILP)	Generic PFD range	Guidance likelihood reduction level (Qualitative)
1	Basic process control system (BPCS)		
	The BPCS control loop / Interlock must use sensors, BPCS input & output (I/O) modules and final elements that are INDEPENDENT of the BPCS control loop / interlock identified as the INITIATING CAUSE	0.1	1 Level
2	Safety Instrumented Function (SIF)		
	Achieves SIL1 PFD _{avg} of 0.1 to 0.01	0.1	1 Level
	Achieves SIL2 PFD _{avg} of 0.01 to 0.001	0.01	2 Level
	Achieves SIL3 PFD _{avg} of 0.001 to 0.0001	0.001	3 Level
3	Alarms with Operator Response		
	(The operator has sufficient time available to respond to the indication of an abnormal condition and complete the required action, and this time is less than the time that it takes for event to become unavoidable)		
	Available to action from control room with Process safety time to action > 1 min	0.1	1 Level
	Manual operator intervention at filed with Process safety time to action > 10 min	0.1	1 Level
4	Routine Operator Surveillance		

No.	Independent Protection Layer (ILP)	Generic PFD range	Guidance likelihood reduction level (Qualitative)
	Sufficient time to indicate the abnormal condition & response and Documented procedures/WI is provide to perform & complete task	0.1	1 Level
5	Adjustable movement-limiting device		
	Device such as strong wire car seal, chain/lock, or adjustable mechanical stop that is intended to prevent operation of a device or movement of an object beyond a defined limit (e.g., strong wire car seal, chain/lock, or adjustable mechanical stop)	0.1	1 Level
6	Pressure Relief Valve (PRV)		
	PRV must be sized to prevent the scenario consequence <i>Note If there is an isolation valve (block valve) upstream or downstream of the PRV without any management system to ensure that valves are in service in their proper positions, then the PFD is taken as 0.1 (likelihood reduction level will be 1 level)</i>	0.01	2 Level
	3 PRV's, where 2 of the 3 PRV's are required to prevent the overpressure and all 3 PRV's are actively connected to the process <u>at all times</u>	0.001	3 Level

No.	Independent Protection Layer (ILP)	Generic PFD range	Guidance likelihood reduction level (Qualitative)
	More than one PRV in clean service <u>is available (Connected to process at all time)</u> to prevent the overpressure consequence. Each PRV listed must be capable of independently relieving the vessel. PRV must be sized to prevent the scenario consequence.	0.001	3 Level
	More than one PRV is available, but more than one is required to relieve the full load. This includes staged release PRV's.	0.1	1 Level
7	Rupture disk		
	Sized for the scenario and failure to open enough at set pressure <i>Note If there is an isolation valve (block valve) upstream or downstream of the PRV without any management system to ensure that valves are in service in their proper positions, then the PFD is taken as 0.1 (likelihood reduction level will be 1 level)</i>	0.01	2 Level
8	Pressure relief valve (PRV) with Rupture disk		
	The rupture disk and relief valve meet the special considerations for the individual devices; Refer to item 7 and item 6.1	0.01	2 Level
9	Conservation vacuum and/or pressure relief vent		

No.	Independent Protection Layer (ILP)	Generic PFD range	Guidance likelihood reduction level (Qualitative)
	Sized for the scenario <i>Note If there is an isolation valve (block valve) upstream or downstream of the PRV without any management system to ensure that valves are in service in their proper positions, then the PFD is taken as 0.1 (likelihood reduction level will be 1 level)</i>	0.01	2 Level
10	Vacuum Breaker		
	Must be designed to prevent the scenario consequence. <i>Note If there is an isolation valve (block valve) upstream or downstream of the vacuum breaker, then the PFD is taken as 0.1, unless there is a management system in place to ensure that valves are returned to service in their proper positions after (likelihood reduction level will be 1 level) maintenance and that they remain in the appropriate state during operation</i>	0.01	2 Level
11	Pressure Reducing Regulator		
	For failure to maintain pressure	0.1	1 Level
12	Restrictive flow orifice		
	Must be designed to prevent the scenario consequence.	0.01	2 Level
13	Check valve		
	Check valve: Must be designed to prevent the scenario consequence	0.1	1 Level

No.	Independent Protection Layer (ILP)	Generic PFD range	Guidance likelihood reduction level (Qualitative)
14	Captive Key/Lock System		
	Captive key/lock system employs mechanical linkages that are released by unique keys to prevent movement of a device	0.01	2 Level
15	Adjustable movement-limiting device		
	An adjustable movement-limiting device, such as A strong wire car seal, chain/lock, or an adjustable mechanical stop, that is intended to prevent operation of a device or movement of an object beyond a defined limit	0.1	1 Level
16	Multiple mechanical pump seal system with seal failure detection and response		
	Multiple mechanical pump seal system includes at least two mechanical seals, with means to detect and indicate a failure of the primary or secondary seal to the operator	0.1	1 Level
17	Mechanical Over-Speed Trip		
	Must be specifically designed for the relevant rotating equipment and appropriately tested as part of the facility's preventive maintenance program	0.1	1 Level

Table 6.5 Index of Indicative guidance for reduction likelihood level of Passive - IPLs

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No.	Independent Protection Layer (ILP)	Generic PFD range	Guidance likelihood reduction level (Qualitative)
1	Flame Arresters		
	End-of-line arrester	0.01	2 Level
	In-line arrester without temperature monitoring	0.1	1 Level
	In-line arrester with temperature monitoring	0.01	2 Level
2	Dikes, Berms, and Bunds		
	Sized for contain the amount of liquid is designed comply with appreciate reference code and standard (e.g., NFPA30, API2510, IP19 etc. depend on type of facility and plant specification/reference standard) <i>Note Using Dikes, Berms, and Bunds for IPLs should be aware. Typically, Dikes, Berms, and Bunds is not considered for Fire and Explosion scenario.</i>	0.01	2 Level
3	Drainage to dikes, berms, and bunds with remote impoundment		
	Sized for contain the amount of liquid and Drainage is designed comply with appreciate reference code and standard (e.g., NFPA30, API2510 etc. depend on type of facility and plant specification/reference standard)	0.01	2 Level
4	Permanent Mechanical Stop That Limits Travel		

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No.	Independent Protection Layer (ILP)	Generic PFD range	Guidance likelihood reduction level (Qualitative)
	Device that limits travel of a component, such as a piston, valve, or machine. - Adjustment or slippage not possible afterwards - Reason for the placement of the mechanical stop is understood and documented	0.01	2 Level
5	Overflow line		
	Overflow line with no impediment to flow. <i>Note IPL Evaluation for this item should be awarded to ensure that this barrier can protected the consequence due to some of scenario this barrier is not appropriate to consider as IPL such when releasing of liquid from overflow line can generate the new hazard such fire, explosion or toxic.</i>	0.001	3 Level
	Overflow Line Containing a Passive Fluid or with a Rupture Disk <i>Note IPL Evaluation for this item should be awarded to ensure that this barrier can protected the consequence due to some of scenario this barrier is not appropriate to consider as IPL such when releasing of liquid from overflow line can generate the new hazard such fire, explosion or toxic.</i>	0.01	2 Level

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6	Inherently Safe Design		
	If properly implemented can significantly reduce the frequency of consequences associated with a scenario <i>Note: the LOPA rules for some companies allow inherently safe design features to eliminate certain scenarios (e.g., vessel design pressure exceeds all possible high-pressure challenges, for such scenrio the potential can be recorded in the way “Not credible scenario since the equipment is designed cover/withstand”).</i>	0.01	2 Level

Table 6.6 Typical safeguards not usually considered IPLs.

No.	Safeguards not usually considered IPLs	Comment
1	Training and certification, Procedures	This faction is used in the assessing the effective for human action but are not the IPLs.
2	Normal testing and inspection	Activities assumed to be in place for all hazard evaluations and form basis for judgment to determine IPLs. Normal testing and inspection affect reliability of protection barrier (affect to probability failure on demand, PFD) of certain IPLs. Lengthening testing/inspection intervals or No testing/inspection intervals may increase PFD of IPL which indicate that the generic failure rate may not appropriate to use in assessment.
3	Communications	Basic assumption is that adequate communications exist in a facility. Poor communications affect PFD of certain IPLs
4	Signs	Signs are not IPLs. Signs may be unclear, obscured, or ignored. Signs may affect PFD of certain IPLs
5	Fire protection	Active fire protection is not often considered an IPL, as it is post event for most scenarios, and its availability and effectiveness may be affected by fire/explosion that it is intended to contain. However, if a company can demonstrate that active fire protection meets IPL for a given scenario it may be used. Fireproof insulation can be used as IPL for some scenarios if it meets API and corporate standards.

